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"""
species_config.py
=================
Loads and validates the species configuration CSV.

The species config CSV controls:
  - Which species to process
  - Their behaviour category (sedentary / mobile / irruptive)
  - Optional per-species overrides of spatial and temporal windows

Expected CSV columns:
  species        (required) Common name, matching exactly your BirdTrack records
  behaviour      (required) One of: sedentary, mobile, irruptive
  radius_km      (optional) Override the default radius for this species
  time_window_hours (optional) Override the default time window

Example rows in species_config.csv:
  species,behaviour,radius_km,time_window_hours,notes
  Marsh Warbler,sedentary,,,"Singing male, typically site-faithful"
  Marsh Sandpiper,mobile,2.0,,"Often moves between nearby pools"
  Waxwing,irruptive,,3,"Flocks, but flag cross-site records"
  Great Snipe,sedentary,0.15,,"Very site-faithful when resting"
  Pallid Swift,mobile,3.0,4,"Hawking over wide area"
"""

import pandas as pd
import sys
from config import BEHAVIOUR_WINDOWS

VALID_BEHAVIOURS = set(BEHAVIOUR_WINDOWS.keys())


def load_species_config(path):
    """
    Load species config CSV and apply per-species window overrides
    back into a merged config DataFrame.
    """
    df = pd.read_csv(path, encoding='utf-8-sig')

    # Normalise column names
    df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')

    if 'species' not in df.columns:
        raise ValueError("species_config CSV must have a 'species' column.")
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    if 'behaviour' not in df.columns:
        raise ValueError("species_config CSV must have a 'behaviour' column.")

    df['species'] = df['species'].str.strip()
    df['behaviour'] = df['behaviour'].str.strip().str.lower()

    # Validate behaviours
    invalid = df[~df['behaviour'].isin(VALID_BEHAVIOURS)]
    if not invalid.empty:
        raise ValueError(
            f"Invalid behaviour values: {invalid['behaviour'].unique().tolist()}.
            f"Must be one of: {sorted(VALID_BEHAVIOURS)}"
        )

    # Apply defaults from BEHAVIOUR_WINDOWS where per-species overrides absent
    def resolve(row, field):
        val = row.get(field)
        if pd.notna(val) and val != '':
            return float(val)
        return BEHAVIOUR_WINDOWS[row['behaviour']][field]

    df['radius_km'] = df.apply(lambda r: resolve(r, 'radius_km'), axis=1)
    df['time_window_hours'] = df.apply(
        lambda r: resolve(r, 'time_window_hours'), axis=1
    )

    print(f"Species config loaded: {len(df)} species")
    for beh in VALID_BEHAVIOURS:
        n = (df['behaviour'] == beh).sum()
        if n:
            print(f"  {beh}: {n} species")

    return df


def generate_example_config(output_path='species_config_example.csv'):
    """
    Write an example species config CSV with common UK rarities
    to help users get started.
    """
    examples = [
        # Sedentary — stays in one location
        ('Marsh Warbler',       'sedentary',  '',    '',   'Singing male at breed
        ('Aquatic Warbler',     'sedentary',  0.15,  4,    'Skulking, very site-f
        ('Great Snipe',         'sedentary',  0.15,  '',   'Resting, rarely moves
        ('Little Crake',        'sedentary',  0.2,   '',   ''),
        ('Baillon\'s Crake',    'sedentary',  0.2,   '',   ''),
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        ('Black-winged Stilt',  'sedentary',  0.3,   '',   'Wades within small we

        # Mobile — moves around but within a local area
        ('Marsh Sandpiper',     'mobile',     2.0,   4,    'Moves between pools a
        ('Pallid Swift',        'mobile',     3.0,   4,    'Hawking over wide are
        ('White-rumped Sandpiper', 'mobile',  1.5,   '',   ''),
        ('Semipalmated Sandpiper', 'mobile',  1.5,   '',   ''),
        ('Lesser Yellowlegs',   'mobile',     2.0,   '',   ''),
        ('Great White Egret',   'mobile',     1.0,   '',   'Moves between wetland
        ('Short-toed Eagle',    'mobile',     5.0,   6,    'Wide-ranging over hil
        ('Pallid Harrier',      'mobile',     3.0,   5,    ''),

        # Irruptive — genuinely could be different individuals
        ('Waxwing',             'irruptive',  0.5,   3,    'Flocks move between b
        ('Two-barred Crossbill','irruptive',  '',    '',   ''),
        ('Parrot Crossbill',    'irruptive',  '',    '',   ''),
        ('Pallas\'s Warbler',   'irruptive',  0.3,   '',   'Arrive in waves, mult
        ('Yellow-browed Warbler','irruptive', 0.3,   '',   ''),
        ('Red-flanked Bluetail','irruptive',  0.3,   '',   ''),
        ('Dusky Warbler',       'irruptive',  0.2,   '',   ''),
        ('Siberian Rubythroat', 'irruptive',  '',    '',   ''),
        ('Taiga Flycatcher',    'irruptive',  '',    '',   ''),
    ]

    config_df = pd.DataFrame(
        examples,
        columns=['species', 'behaviour', 'radius_km', 'time_window_hours', 'notes
    )
    config_df.to_csv(output_path, index=False)
    print(f"Example species config written to: {output_path}")
    return config_df


if __name__ == '__main__':
    generate_example_config()
    print("\nEdit species_config_example.csv, then rename to species_config.csv")
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